

MEDS Lab Module 4

Section 7: Decoders, Encoders & One-Hot Logic

Task 6: 3-to-8 Decoder from 2-to-4 Decoders

July 23, 2026

1. EDA Playground Link

The complete design and testbench code for the standalone 2-to-4 decoder and the instantiated 3-to-8 decoder can be found at the following shareable link:

- <https://edaplayground.com/x/kTZX>

2. Schematic Diagram

Below is the top-level block diagram illustrating the logic used to construct the 3-to-8 decoder from two smaller 2-to-4 decoders.

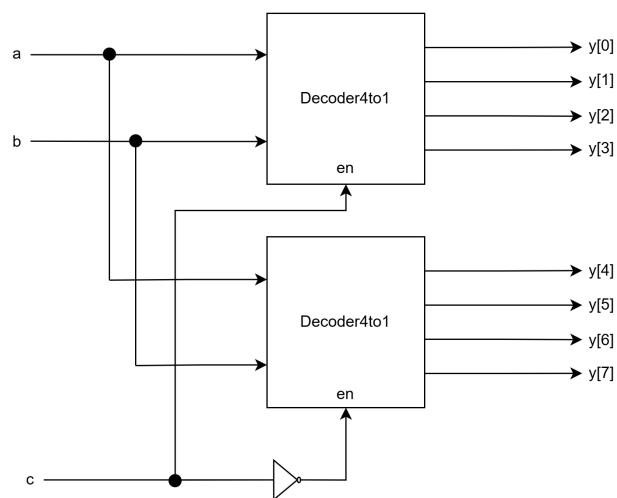
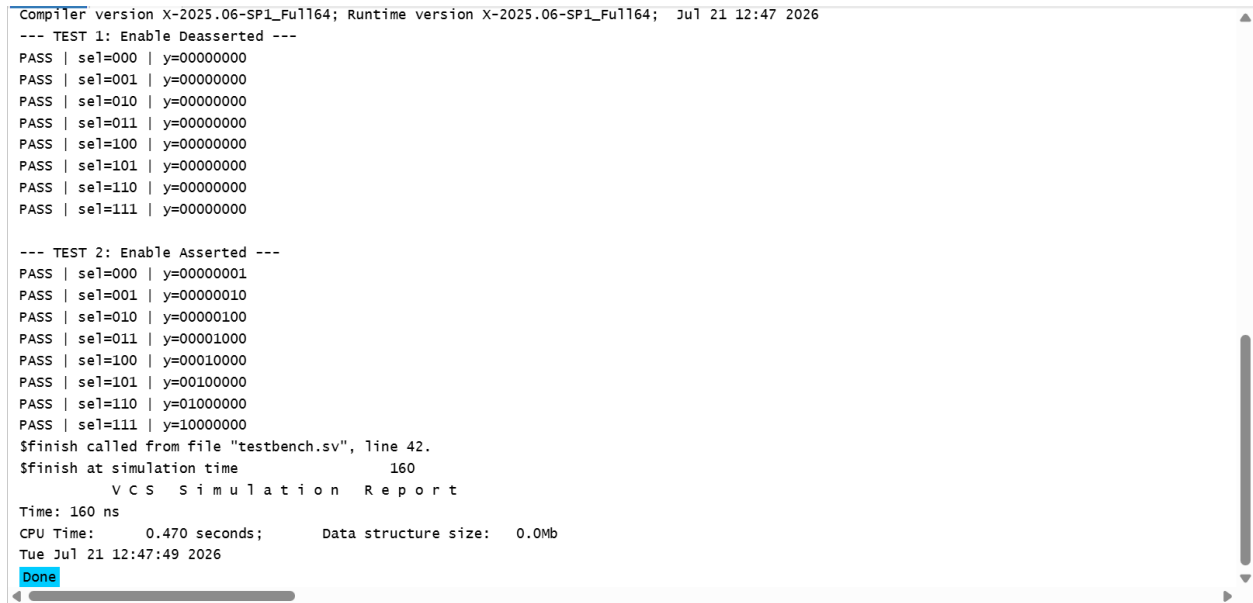


Figure 1: Structural composition of a 3-to-8 decoder using two 2-to-4 decoders.

3. Terminal Output Screenshot

The following screenshot displays the simulation log. It demonstrates that the testbench exhaustively sweeps all 8 select values with the enable asserted, and confirms that all outputs correctly remain 0 when the enable is deasserted.



```
Compiler version X-2025.06-SP1_Full164; Runtime version X-2025.06-SP1_Full164; Jul 21 12:47 2026
--- TEST 1: Enable Deasserted ---
PASS | sel=000 | y=00000000
PASS | sel=001 | y=00000000
PASS | sel=010 | y=00000000
PASS | sel=011 | y=00000000
PASS | sel=100 | y=00000000
PASS | sel=101 | y=00000000
PASS | sel=110 | y=00000000
PASS | sel=111 | y=00000000

--- TEST 2: Enable Asserted ---
PASS | sel=000 | y=00000001
PASS | sel=001 | y=00000010
PASS | sel=010 | y=00000100
PASS | sel=011 | y=00001000
PASS | sel=100 | y=00010000
PASS | sel=101 | y=00100000
PASS | sel=110 | y=01000000
PASS | sel=111 | y=10000000
$finish called from file "testbench.sv", line 42.
$finish at simulation time          160
      V C S   S i m u l a t i o n   R e p o r t
Time: 160 ns
CPU Time:      0.470 seconds;      Data structure size:  0.0Mb
Tue Jul 21 12:47:49 2026
Done
```

Figure 2: Simulation log output from EDA Playground verifying the 3-to-8 Decoder.